

HAMILTONIAN CYCLES ON SQUARE GRIDS WITH A CENTRAL 2×2 HOLE

JASON ROELL

ABSTRACT. We enumerate undirected Hamiltonian cycles on the $2n \times 2n$ lattice grid with its four central vertices deleted, for $1 \leq n \leq 10$. Two frontier implementations with different connectivity encodings give identical exact counts. Half-board and quarter-board reductions count cycles fixed by square symmetries, yielding geometric orbit counts and exact stabilizer classes. We prove that quarter-turn symmetry is impossible when n is odd. A separate edge-orbit search verifies the symmetry counts through the 8×8 case, and additionally verifies quarter-turn and both-axis counts through 12×12 . The source code, tables, resource measurements, and verification procedures are public. The finite-size comparisons are descriptive and do not establish a new limiting growth constant or exponent.

1. DEFINITION AND EXACT SEQUENCE

Let G_n have vertices (x, y) with $1 \leq x, y \leq 2n$, except for (n, n) , $(n, n + 1)$, $(n + 1, n)$, and $(n + 1, n + 1)$. Edges join lattice neighbors. For $n \geq 2$, the graph has $4n^2 - 4$ vertices and $8n^2 - 4n - 12$ edges. Let H_n count its Hamiltonian cycles as undirected edge sets, with neither a starting vertex nor a direction distinguished. Rotated and reflected edge sets are initially distinct. Set $H_1 = 0$, since G_1 is empty. The first nonempty case G_2 is a single 12-cycle.

n	$ V(G_n) $	H_n
1	0	0
2	12	1
3	32	14
4	60	164102
5	96	9684216390
6	140	27867828294632440
7	192	1167509412814480584454378
8	252	1517021583328338160591145724550850
9	320	35604867201983888335090959513825934093936960
10	396	22194400531163887940263507134424320689889325050239815212

TABLE 1. Undirected Hamiltonian cycles on the centrally punctured grid.

The computation gives the complete table through $n = 10$ shown in Table 1. In particular,

$$H_8 = 1517021583328338160591145724550850.$$

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The number O_n of geometric orbits under the eight symmetries of the square is calculated in Section 5. The code and data are at

<https://github.com/jroell/punctured-grid-cycles>.

The counting programs and complete data are fixed at commit `2d0d15b`. The revised manuscript and its build instructions are in the same repository.

Connectivity transfer methods are established techniques for Hamiltonian grid enumeration. The present calculation adapts them to this central puncture. The relation to earlier work is discussed in Section 8.

2. FRONTIER RECURRENCE AND CORRECTNESS

Process lattice positions in row-major order. A frontier separates processed positions from unprocessed positions and has at most $w + 1$ slots for a board of width w . Each slot is empty or carries a selected edge to the unprocessed side. Every processed vertex has its required degree, and every component built so far is a path. The state records which occupied slots are the two ends of each path.

In the primary implementation, planarity makes the endpoint pairing noncrossing. Empty slots are encoded by 0, and the two ends of each pair by opening and closing parentheses, represented by 1 and 2. A word uses $2(w + 1)$ bits. The checker stores each partner index explicitly, without parenthesis matching. Both implementations aggregate equal states in hash maps. Empty slots and noncrossing pairs form Motzkin words, giving an upper bound of M_{w+1} possible states. At width 16 this bound is $M_{17} = 2356779$; at width 20 it is $M_{21} = 142547559$. Only reachable states are created.

2.1. Local transitions. At a remaining vertex, the numbers of selected incoming and outgoing edges sum to two. There are five cases.

- (1) With no incoming edge, select both outgoing edges, right and down, if both neighbors exist, and create a paired path.
- (2) With one incoming edge, select one available outgoing edge and move the incoming path endpoint to it.
- (3) With two incoming edges from different paths, select no outgoing edge, join the paths, and pair their other endpoints.
- (4) With two incoming edges from the same path, accept only at the last remaining vertex, with no other occupied frontier slots.
- (5) At a deleted position, permit no selected incident edges. Pass the state through only when both local slots are empty.

If $D_t(s)$ counts partial edge sets producing state s after t positions, then

$$D_{t+1}(s') = \sum_s \sum_{\substack{\text{legal local choices} \\ s \rightarrow s'}} D_t(s).$$

Initialize the empty state with weight one. At row boundaries, shift the contour slots. At the final vertex, accumulate only a legal single-cycle closure. Each local update has at most two choices; locating a matching parenthesis can scan $O(w)$ slots.

2.2. Correctness. Induct on the processing order. Every legal transition gives the newly processed vertex degree two and preserves the path interpretation of the frontier. Joining two ends of the same path is the only way to create a closed component. Rejecting that transition before the final closure therefore excludes every disconnected 2-factor. Conversely, restricting a Hamiltonian cycle to successive processed regions produces exactly these legal states and choices. Every edge is decided once, at its earlier endpoint. Thus every accepted edge set has one construction history, and no division by the cycle length or by two is required.

The implementations use checked unsigned 256-bit counters, represented by four 64-bit limbs. Addition aborts on overflow instead of wrapping. No overflow occurred in the recorded runs. Accepted dimensions are at most 24, keeping both state encodings within their supported widths. The verified main tables in this note cover $n \leq 10$.

3. HOLE HANDLING, PRUNING, AND STATE COUNTS

A removed vertex contributes only an empty local transition. Edges pointing into the hole are disabled before selection. The scan still crosses the entire rectangular board. Empty slots at the hole do not split the state into independent left and right problems: paths on opposite sides may already be connected through the processed region. Their pairing is retained globally.

The hole creates an inner face, but does not invalidate noncrossing pairings on the outer scan contour. Disjoint planar paths with endpoints on that contour cannot realize an interlacing pairing. No winding label is needed for the raw count, because the pairing retains the connectivity relevant to subsequent extensions.

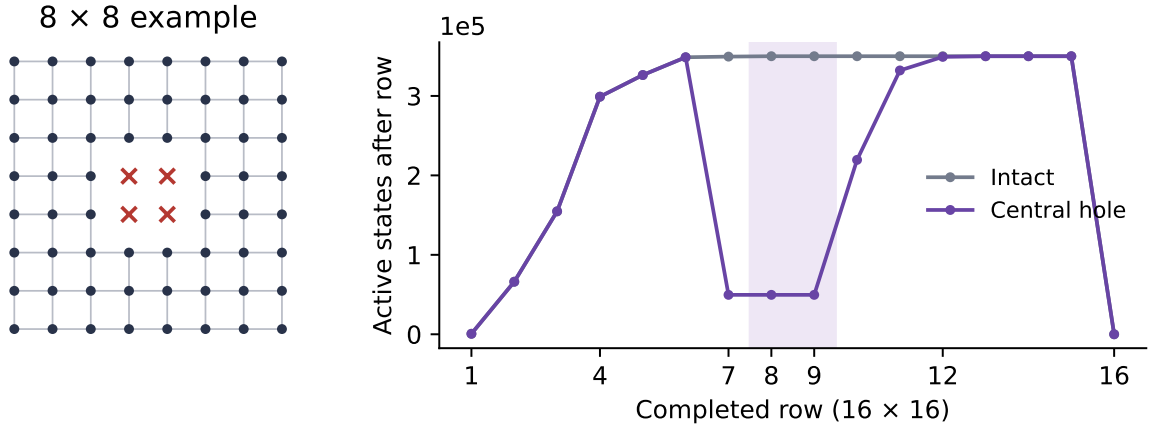


FIGURE 1. Deleted vertices in the 8×8 example, and row-end state counts at side 16. Shading marks the rows with deleted vertices. The reduction starts one row earlier because edges into the hole are disabled. The maximum over all vertex transitions exceeds the row-end maximum.

At side 16, the peak of 677,909 active states first occurs after row 13, column 13. The row-end maximum is 349998. The two hash maps coexist during a transition; process RSS includes both maps, their buckets, and allocator overhead. The recorded run took 11.01 seconds and used 315.8 MiB peak RSS. Measurements are single observations on an Apple M4 Max with 128 GiB RAM, using Apple clang 21.0.0 with `-O3 -std=c++17`. They are not

$2n$	Hole peak	Intact peak	Seconds	RSS (MiB)
4	1	11	0.000010	1.719
6	40	59	0.000040	1.750
8	339	341	0.000233	1.891
10	2,120	2,120	0.002916	2.578
12	13,943	13,943	0.035642	5.422
14	95,660	95,660	0.476549	24.375
16	677,909	677,909	11.006828	315.797
18	4,928,252	4,928,252	19.928776	1233.266
20	36,572,939	36,572,939	205.434662	9856.422

TABLE 2. Primary-counter peak states for punctured and intact boards. Time and memory refer to punctured boards. Sides through 16 use standard hash maps; larger sides use Boost flat maps, so timings are not directly comparable.

repeated, isolated performance benchmarks. The larger runs use Boost 1.92.0 flat hash maps for aggregation. The explicit-partner counter also supports local updates to partner fields rather than decoding and re-encoding the whole frontier at every vertex. These changes preserve the state definition and recurrence. All 46 established raw and symmetry runs were rechecked with the flat-map backend, including peak states. The punctured side-18 count was also checked with both original standard-map implementations. Some extension and validation runs overlapped on the workstation; their timings are not isolated comparisons.

Pruning uses degree constraints, missing-neighbor constraints, and rejection of premature cycles. The raw counter does not identify states under rotations or reflections, and does not perform future-reachability pruning. Symmetries are counted separately by the reductions below.

4. FIXED CYCLES ON SMALLER BOARDS

Write R_2 for the number of full-board cycles fixed by 180° rotation, R_4 for those fixed by 90° rotation, F for those fixed by one specified axial reflection, and B for those fixed by both axial reflections. These counts depend on n , suppressed in the notation. The symmetry counter uses explicit endpoint partners and temporary boundary stubs, and never closes a path inside the smaller board.

4.1. Half board: F and R_2 . Take rows 1 through n , with the two central vertices deleted from the last row. A reflection-fixed spanning cycle must cross the reflection axis. Every crossing edge is fixed setwise. The induced order-two cycle automorphism is either a reflection or a half-shift. A half-shift fixes no edge setwise, so the crossing rules it out. Thus the automorphism is a reflection with no fixed vertices and exactly two fixed edges. These are exactly the two crossings. The retained half is a Hamiltonian path with both endpoints at allowed positions on the cut. Counting final states with exactly two paired stubs gives F .

The half-turn has its center at a lattice face center and fixes no vertex or edge setwise. It therefore acts freely on vertices and edges of the cycle. Identify omitted vertices with their rotated representatives. On the cut, pair column x with column $2n + 1 - x$. Selected seam edges complete the half-board paths into a quotient cycle. Require one quotient component and an odd number of seam edges. Each seam changes the copy index by one modulo two,

so odd parity makes the lift one cycle instead of two. This gives R_2 without constructing full-board cycles.

4.2. Quarter board: R_4 and B . For a quarter-turn, use the $n \times n$ quadrant with corner (n, n) deleted. Add seam edges joining (n, k) to (k, n) , for $1 \leq k < n$. A connected quotient cycle lifts to one cycle on four copies precisely when its net change of copy index is coprime to four. Each seam contributes $+1$ or -1 . Thus the condition is equivalent to an odd number of seam edges. Final states are tested for this parity and a single quotient component. This is the punctured analogue of the quadrant construction used by Wynn [2] for intact grids.

Quarter board: 4×4 minus one corner

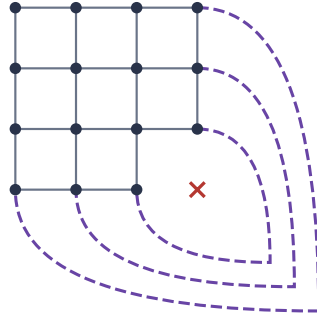


FIGURE 2. Quarter-board quotient for side 8. Dashed curves are additional seam edges, not original grid edges. The corner $(4, 4)$ is deleted.

If a full cycle is fixed by both axial reflections, it crosses each axis exactly twice. Cutting at those four edges leaves one spanning path in each quadrant. Its endpoints lie on the two inner sides. Counting quarter-board states with exactly two paired stubs, one on each inner side, gives B . Reflection reconstructs the full cycle uniquely.

5. PARITY, DIAGONAL REFLECTIONS, AND BURNSIDE'S LEMMA

Proposition 1. *If $n \geq 3$ is odd, then G_n has no Hamiltonian cycle fixed by a quarter-turn. Equivalently, $R_4 = 0$ for odd n .*

Proof. Color (x, y) by the parity of $x + y$. A quarter-turn sends it to $(2n + 1 - y, x)$ and exchanges the colors. Suppose a Hamiltonian cycle C is fixed by this rotation. Its length is $N = 4n^2 - 4$. The induced automorphism of C has order four, since C spans the graph and the quarter-turn acts faithfully on its vertices. Every order-four automorphism of an undirected cycle is a cyclic shift by $N/4$ or $3N/4$ positions. These are $n^2 - 1$ and $3(n^2 - 1)$ positions. If n is odd, both shifts are even. Since colors alternate around C , such a shift preserves color, contradicting the color exchange of the quarter-turn. $\square$

The empty case $n = 1$ also has $R_4 = 0$ by convention. The deletion reverses the corresponding parity obstruction for intact grids [2].

Proposition 2. *For $n \geq 3$, neither diagonal reflection fixes a Hamiltonian cycle of G_n .*

Proof. Each diagonal reflection fixes $2n-2 \geq 4$ remaining vertices. If it preserved a Hamiltonian cycle, its restriction would be a nonidentity cycle automorphism with at least four fixed vertices. A nonidentity cycle automorphism fixes at most two vertices, a contradiction. $\square$

Let O_n count geometric orbits of Hamiltonian-cycle edge sets under D_4 . For $n \geq 3$, Proposition 2 and Burnside's lemma give

$$(1) \quad O_n = \frac{H_n + R_2 + 2R_4 + 2F}{8}.$$

The two quarter-turns have the same fixed set, and the two axial reflections are conjugate. For $n = 2$, add the two diagonal-reflection contributions of one before dividing by eight; the unique 12-cycle is fixed by all of D_4 . Set $O_1 = 0$.

n	R_2	R_4	F	B
2	1	1	1	1
3	2	0	4	2
4	398	18	436	20
5	39598	0	58198	138
6	155368312	8650	96650662	6406
7	480564699890	0	308860488706	201338
8	34116385160498522	106253220	10202488985967222	41184930
9	2811552194264884895872	0	740001412241179546444	6067605252
10	3914442037913372967313267520	31064776598948	515530816787170934773269074	5594443405532

TABLE 3. Fixed edge-set counts for the punctured grids. F refers to one specified axial reflection; B requires both.

n	O_n
1	0
2	1
3	3
4	20676
5	1210546548
6	3483478580414922
7	145938676601947358766460
8	189627697916042276889063633686282
9	4450608400247986041886906383605585167240715
10	2774300066395485992532938392421228045172137752081602347

TABLE 4. Geometric orbits under all eight symmetries of the square.

In particular,

$$O_8 = 189627697916042276889063633686282.$$

These are geometric equivalence classes, not classes under arbitrary abstract graph isomorphism. Every computed Burnside numerator is divisible by eight. This is a consistency check; the separate small-case edge-orbit search provides a different enumeration check.

6. EXACT STABILIZER CLASSES

For $n \geq 3$, a cycle cannot have both quarter-turn and axial-reflection symmetry, since these would generate a forbidden diagonal reflection. The five possible exact stabilizers have the orbit counts in Table 5.

Exact stabilizer	Orbit size	Number of orbits
Identity only	8	$(H_n - R_2 - 2F + 2B)/8$
One axial reflection	4	$(F - B)/2$
Half-turn only	4	$(R_2 - R_4 - B)/4$
Both axes and half-turn	2	$B/2$
Quarter-turn rotations	2	$R_4/2$

TABLE 5. Exact stabilizer decomposition for $n \geq 3$.

To derive these formulas, first note that B and R_4 count disjoint sets with order-four stabilizers, each with two members per orbit. Subtracting these sets from R_2 leaves the half-turn-only edge sets in orbits of size four. Each one-reflection orbit has two members fixed by the specified axis, which gives $(F - B)/2$. Subtracting all nontrivial stabilizers from H_n gives the identity-only count. This bookkeeping follows the corresponding intact-grid analysis of Wynn [2].

Exact stabilizer	Number of orbits
Identity only	189627697916042263258722834310968
One axial reflection	5101244472391146
180° rotation only	8529096253265093
Both axial reflections	20592465
Quarter-turn rotations	53126610
Full D_4	0
Total	189627697916042276889063633686282

TABLE 6. Exact symmetry classes for the 16×16 punctured grid.

The class counts are nonnegative integers at every computed size. Their sum is O_n , and weighting them by the orbit sizes 8, 4, 4, 2, 2 recovers H_n . For $n = 2$, there is instead one orbit with full D_4 stabilizer and orbit size one.

7. FINITE-SIZE COMPARISON WITH INTACT GRIDS

Let C_n count undirected Hamiltonian cycles on the intact $2n \times 2n$ grid, as in OEIS A003763 [1]. Both full counters reproduce its values for $n \leq 10$; the OEIS b-file contains terms through $n = 13$. Define

$$Q_n = \frac{H_n}{C_n}, \quad c_n = C_n^{1/(4n^2)}, \quad h_n = H_n^{1/(4n^2-4)}.$$

The same-footprint ratio Q_n is not a probability of surviving vertex deletion: deleting vertices from an intact Hamiltonian cycle generally does not leave a Hamiltonian cycle. The per-vertex quantities c_n and h_n are finite-size growth estimators. Both vertex count and boundary corrections matter in interpreting them.

For context, Jacobsen [3, Eq. (4.5)] reports the numerical estimate $\mu = 1.472801 \pm 0.00001$ for the square-lattice Hamiltonian-walk connective constant, attributing it to Jacobsen and Kondev; his own circuit-and-walk analysis gives 1.473 ± 0.001 in Eq. (4.4). This is a numerical benchmark, not an exact constant or a proven limit for the punctured sequence considered here.

n	H_n/C_n	$\log Q_n$	c_n	h_n
3	0.013059701	-4.338224012	1.213869718	1.085966682
4	0.035377668	-3.341674516	1.271048904	1.221570580
5	0.020725521	-3.876389441	1.308264601	1.270637022
6	0.025894009	-3.653743666	1.334201899	1.310584485
7	0.020801394	-3.872735292	1.353235262	1.334597523
8	0.023026163	-3.771124175	1.367762905	1.354161936
9	0.020534244	-3.885661351	1.379198061	1.368038963
10	0.021749399	-3.828169175	1.388423611	1.379631942

TABLE 7. Finite-size ratios and per-vertex growth estimators.

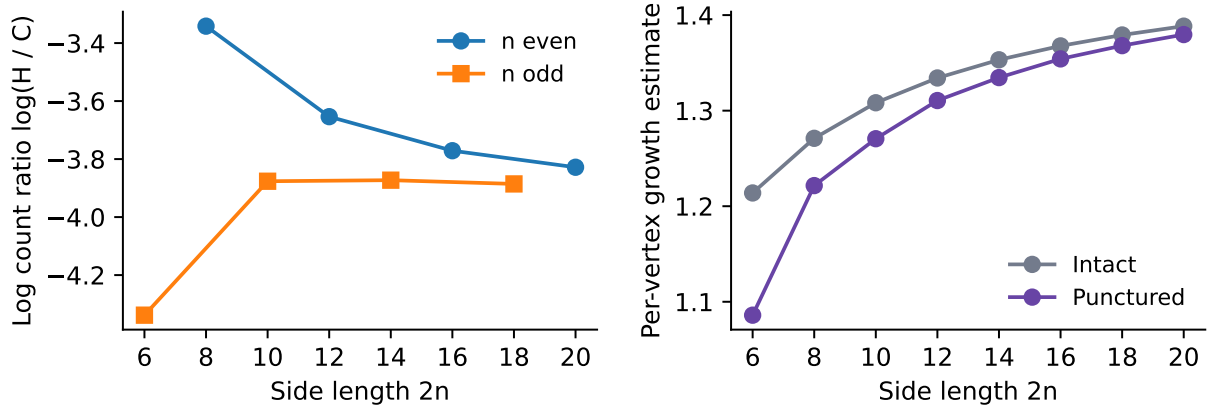


FIGURE 3. Exact-count comparisons. Lines connect computed values only; no asymptotic curve is fitted.

Over $3 \leq n \leq 8$, the even- n ratios decrease and the odd- n ratios increase. The extension to $n = 9$ reverses the latter trend: $Q_9 \approx 0.020534244$, below $Q_7 \approx 0.020801394$. Thus monotone convergence of the odd subsequence is not supported even by this modest extension. The hole deletes two vertices of each checkerboard color for every n , so color imbalance alone cannot explain the parity split. At $n = 8$, the punctured count is about 2.3026% of the intact count for the same outer board. The available finite-size values do not identify an asymptotic exponent or distinguish a constant prefactor from a slowly changing correction. The log ratio $\log Q_n$ measures the change in the logarithm of the number of configurations caused by the defect. Its negative can be viewed as a dimensionless defect free-energy cost. This observable removes the common outer footprint without introducing different per-vertex normalizations. Figure 3 plots $\log Q_n$ separately by parity. Multiplying Q_n by the numerical benchmark μ^4 would compensate heuristically for four missing vertices; on the log scale this

only adds $4 \log \mu$. It does not establish a limiting defect cost. A constant limit, logarithmic correction, and parity-dependent corrections remain hypotheses to test with larger boards. No gluing argument proving equality of the bulk coefficients is supplied here.

8. REPRODUCIBILITY AND RELATED WORK

The public repository <https://github.com/jroell/punctured-grid-cycles> contains the source code, recorded data, and manuscript build instructions. The complete exact-count reference is commit 2d0d15b.

8.1. Verification. The parenthesis and direct-partner frontier counters agree on all intact and punctured boards at every even side from 2 through 20. They share the integer type and the row-by-row decomposition, so these are separately encoded cross-checks, not wholly independent mathematical methods. Direct path enumeration checks the complete cycle sets at punctured sides 4 and 6, including all eight symmetry actions. A different Python checker branches on whole edge orbits and independently verifies R_2, R_4, F, B through punctured side 8. Additional runs of that checker verify R_4 and B at sides 10 and 12. Larger fixed-set counts still rely on the quotient implementation and consistency checks, not an independent second enumeration. The intact results agree with OEIS A003763 [1].

From a repository checkout, run

```
make all
make benchmark
make test
make flat BOOST_CPPFLAGS=-I/opt/homebrew/include
python3 scripts/verify_backends.py
python3 scripts/verify_more_symmetry.py
python3 scripts/extend.py --backend flat
python3 scripts/extend_symmetry.py
python3 scripts/verify_extension.py
uv sync --group paper
uv run --group paper python scripts/build_paper.py
```

The optional Boost include path shown is for Homebrew on Apple Silicon; omit it when Boost headers are on the default compiler include path. The counters need a C++17 compiler supporting `unsigned __int128` and Python 3.11 or later. The manuscript build additionally uses Tectonic and Pandoc; the Python document dependencies are pinned in `uv.lock`. The original full-board benchmark uses a 300-second timeout per run. The symmetry counter stops after 1800 seconds or fifty million active states. The original symmetry benchmark additionally has a 260-second subprocess timeout. Larger raw runs have a 1800-second timeout and a monitored 48 GiB RSS ceiling; the RSS monitor polls every two seconds. Each small edge-orbit check has a 60-second limit. All jobs run finite case lists. Counts are stored as decimal strings in JSON to avoid floating-point rounding. Timings and memory use may differ by platform.

8.2. Prior work and scope. Wynn [2] counts symmetry classes for intact square grids and uses quadrant quotients for quarter-turn symmetry. Jacobsen [3] enumerates circuits, walks, and chains by transfer methods. Blanco and Zeilberger [4] give a recent fixed-width generating-function implementation. The present note applies these ideas to a specified central puncture and records a reproducible table through side 20.

OEIS searches for consecutive terms of both H_n and O_n returned no match on September 26, 2026. These searches and the literature check are not exhaustive and do not establish priority. No OEIS identifier or preprint identifier is asserted before it has been assigned.

REFERENCES

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